Q1, Force & pressure :

a) Define atmospheric Pressure and state its SI unit.

The pressure exerted by the weight of air in the atmosphere is called atmospheric Pressure SI unit : Pascal (Pa)

b) Explain why sharp cutting edge tools exert significally higher Pressure that blunt ones, referencing the formula $P = F/A$.

Q2, Sound wave Terminology :

Define the following technical parameters of a acoustic wave.

a, Amplitude and its relationship with the loudness of sound

Amplitude is the maximum displacement of vibrating Particle from its mean positive. Greater amplitude Produces louder noise.

b, Frequency and its SI unit. explaining how it determines the pitch (shrillness).

Frequency is the number of vibrations / oscillations per second.

SI-unit : Hertz (Hz)

Higher frequency - higher pitch / shriller sound.

Q3, Friction Types:

Arrange static friction, sliding friction and rolling friction in increasing order of their magnitude. Define interlocking of irregularities and explain how lubricants reduce friction.

Increasing order:

Rolling friction < sliding friction < static friction

Interlocking of irregularities:

The tiny irregularities on two surfaces get locked with each other. opposing motion.

Lubricants:

Lubricants such as oil and grease form a thin layer between surfaces, reducing direct contact and therefore reducing friction.

Q4, Chemical Effects of Electric Current.

Define electroplating and electrodes. Name the electrolyte and the cathode material used when coating an iron object with copper.

Electroplating: The process of depositing a thin layer of one material over another using electric current is called electroplating.

Electrodes: conducting plates / rods through which electric current enters or leaves the electrolyte.

For coating iron with copper:

Electrolyte : copper sulphate ($CuSO_4$) solution.

cathode: iron object

Anode: copper plate

Q5, Optics & Reflection:

a, State the two fundamental laws of Reflection.

Laws of reflection:

1, Angle of incidence = angle of reflection

2, The incident ray, reflected ray and normal all lie in the same plane.

b, If a ray of light strikes a plane mirror making an angle of 35° with the glancing surface (mirror surface), calculate the angle of incidence (θ_i) and angle of reflection (θ_r).

$i + r = ?$

Since the angle is given with the mirror surface then,

$i = 90° - \text{given angle}$ & $r = i$

Q6, Electrostatics & Natural phenomena:

a, Differentiate btw a conductor and an insulator based on free electron mobility.

conductor	insulator
free electrons can move easily	Electrons cannot move freely.
Example: copper	Example: Rubber

b) Explain the working mechanism of an electroscope in detecting charges via the process of charge transfer and repulsion.

When a charged object touches or comes near the electroscope, charge is transferred/induced. The leaves acquire similar charges and repel each other, showing the presence of electric charge.

Q7, Acoustics & Noise Pollution:

a, difference btw audible & infrasonic/ultrasonic frequency bands for human hearing. State the exact frequency spectrum for human audibility.

* Infrasonic: Below 20 Hz
* Audible: 20 Hz - 20,000 Hz
* Ultrasonic: Above 20,000 Hz

Thus, humans can normally hear frequencies from:

20 Hz to 20,000 Hz